Unified Review: Multitone SQR Control and Holographic Cluster-State Synthesis in Dispersive cQED

A Critical Audit, Rerun, and Synthesis of Nine Prior Studies

Critical Review — March 28, 2026

Contents

1 Overview and Scientific Questions

This document audits, reruns, and unifies the nine most recent studies in this research repository, all of which address one or both of the following related questions:

Scientific Question 1 (SQR): Can a parameterized multitone qubit-drive waveform simultaneously realize a *strict ideal x-axis selective qubit rotation* (SQR) $U_n = \exp(-i\theta_n X/2)$ on each addressed Fock manifold $n \in \{0, 1, \ldots, N_{\text{active}}-1\}$ of a dispersive transmon-cavity system, with no residual per-manifold conditional phase, to a joint process fidelity exceeding 0.99? And if so, under what conditions (pulse duration, number of addressed manifolds, Hamiltonian model) does this succeed, and where does it break down?

Scientific Question 2 (Cluster State): Can a structured native-gate sequence (D + R + SQR or D + R + CPSQR) implement the restricted ground-sector holographic cluster-state transfer $U_{\text{joint}}(|g\rangle \otimes |\psi\rangle) \approx |g\rangle \otimes H_c|\psi\rangle$ on $\{|g, 0\rangle, |g, 1\rangle\}$ to fidelity exceeding 0.99, and how does the minimum circuit depth compare to the unconstrained GRAPE benchmark?

The nine studies were carried out autonomously by a prior agent. This review reexamines every study skeptically, checks all key claims against the saved data and against rerun simulations, identifies weaknesses and inconsistencies, and produces a unified set of evidence-backed conclusions.

2 Physical Model and Shared Conventions

2.1 Hamiltonian

All nine studies share the same dispersive transmon-cavity Hamiltonian:

$$H = \omega_c a^\dagger a + \omega_q |e\rangle\langle e| + \frac{\alpha}{2}|e\rangle\langle e|(|e\rangle\langle e| - 1) - \chi a^\dagger a |e\rangle\langle e| - \frac{\chi'}{2}(a^\dagger a)^2 |e\rangle\langle e| + K(a^\dagger a)^2 \tag{1}$$

with parameters:

Parameter	Symbol	Value
Qubit frequency	$\omega_q/2\pi$	6.150 GHz
Cavity frequency	$\omega_c/2\pi$	5.241 GHz
Transmon anharmonicity	$\alpha/2\pi$	−255 MHz
Dispersive shift	$\chi/2\pi$	−2.84 MHz
Second-order dispersive shift	$\chi'/2\pi$	−21 kHz
Cavity Kerr	K	0 (set to zero in all studies)
Transmon levels retained	N_{tr}	2
Cavity levels	N_{cav}	4–12 (study dependent)

The SQR studies use two model variants: `chi_only` (sets $\chi' = 0$) and `chi_plus_chiprime` (full model). The cluster-state study uses the full model with an additional Kerr term $K/2\pi = -28$ kHz for the GRAPE benchmark only.

2.2 SQR Target Definition

A **strict ideal SQR** on N_{active} addressed Fock manifolds requires:

$$U_{\text{joint}}\big|_{n\in\text{addr}} = R_x(\theta_n) \equiv e^{-i\theta_n X/2}, \qquad U_{\text{joint}}\big|_{n\notin\text{addr}} = \mathbf{1} \tag{2}$$

with *identical rotation angle* θ_n *for all* n and *zero residual Z phase*. The minimum free parameter count for this target is $2N_{\text{active}}$ real numbers (one θ_n and one phase-lock constraint per manifold).

A **relaxed CPSQR** (conditional phase SQR) allows arbitrary per-manifold Z phases φ_n:

$$U_{\text{joint}}\big|_n = e^{i\varphi_n Z/2} R_x(\theta_n) \tag{3}$$

The CPSQR target is substantially easier because the optimizer is not required to suppress conditional phase accumulation.

2.3 Metric Hierarchy

Multiple metrics appear across the nine studies, and their inconsistent use is a major source of confusion. In order of increasing stringency:

1. **Reduced/restricted process fidelity** — fidelity of the per-manifold reduced effective-qubit unitary, ignoring cavity-mode correlations. *Most optimistic; can be high even when the full joint operator is wrong.*

2. **Single-ground-state probe fidelity** — state fidelity on $|g\rangle$ input only. Misses all coherence errors.

3. **Spanning-quartet probe fidelity** — mean state fidelity over $\{|g\rangle, |e\rangle, |+x\rangle, |+y\rangle\}$ inputs per manifold. Sufficient to certify a per-manifold unitary up to global phase.

4. **Full joint process fidelity** — process fidelity of the complete joint qubit-cavity unitary on all addressed and unaddressed manifolds simultaneously. *Most stringent; the definitive metric.*

A key finding from Study 4 (multi-input validation) is that *metrics 1 and 2 can yield high values while metric 4 is essentially zero.* The definitive metric throughout this review is the full joint process fidelity.

3 The Nine Studies: Goals, Methods, and Weaknesses

#	Study	Primary goal	Key weakness identified in this review
1	`corrected_sqr_conditional_rotation_metric`	Establish correct qubit-effective metric for SQR optimization	Only thoroughly validates $N_{\text{active}} = 1$; multi-manifold extension not demonstrated
2	`parameterized_waveform_residual_z_cancellation`	Test richer waveform families for Z-error suppression	Reports restricted process fidelity; multi-input validation (Study 4) shows the baseline result is essentially broken ($F_{\text{joint}} = 0.008$)
3	`ideal_sqr_direct_vs_echoed_multitone`	Formulate strict ideal x-axis SQR; direct vs echoed	Best direct case has `optimizer_success=False`; echoed does not demonstrably improve strict metric
4	`native_multitone_sqr_fixed_multi_input_validation`	Validate single-input artifacts from Studies 1–3 on multi-input probe ladder	Reveals that Studies 2 and 3 artifacts fail on superposition probes
5	`multitone_sqr_arbitrary_block_conditional_rotations`	Broaden to arbitrary SU(2) block targets; direct vs echoed	Mean process fidelity 0.35 (direct) and 0.22 (echoed); echoed is *worse* on average
6	`strong_validation_arbitrary_fock_conditional_rotations`	5 waveform families; strict and relaxed CPSQR evaluation	Higher-expressivity benchmark yields zero gain; reason not investigated
7	`native_rich_multitone_sqr_cpsqr_feasibility`	Most comprehensive: 10 families, 3-stage screen, both models	High quartet validation on separate manifolds does not certify joint operator (one case: quartet = 0.9999997, joint = 0.868)
8	`the_definitive_ideal_sqr_gate_study`	Aggregate and synthesize all prior SQR results	Aggregation-only pass; no new optimizations run; practical ranking uses decoherence proxy, not actual drift sweeps
9	`cluster_state_holographic`	Unified cluster-state ground-sector synthesis	Old full-target winners completely fail for ground-sector task (ancilla excitation ≈ 1); no open-system validation for ground-sector branch yet

3.1 Study 1: Corrected Metric (`corrected_sqr_conditioned_rotation_metric`)

Goal. Establish that the correct objective for SQR optimization is the reduced effective-unitary weighted process fidelity, not raw state fidelity. Verify the phase convention used by `cqed_sim`'s multitone ansatz (specifically, confirm that ϕ is an axis-angle, not a quadrature offset) and demonstrate analytic and gradient-based correction methods.

Methods. A 4-level target profile with varying θ_n and ϕ_n per manifold. Four optimization methods compared: baseline (no correction), analytic warm-start, state-based optimization, and unitary-based optimization.

Results. For $N_{\text{active}} = 1$, the unitary-optimized case achieves $F_{\text{joint}} = 0.9999999$ and zero residual Z or transverse error. For $N_{\text{active}} = 4$, the baseline fidelity is poor but the corrected metric guides the optimizer to better solutions.

Weakness. The study validates the metric and convention definitively for $N_{\text{active}} = 1$ but the multi-manifold cases ($N_{\text{active}} = 4$) are described but not independently validated. The study summary shows per-manifold convergence checks but not the full joint process fidelity at $N_{\text{active}} = 4$. Study 4 multi-input validation confirms that the corrected-metric unitary-optimized case achieves 1.0000 spanning-quartet fidelity, which is the strongest per-manifold validation available short of a full GRAPE comparison.

3.2 Study 2: Parameterized Waveform (`parameterized_waveform_residual_z_cancellation`)

Goal. Pilot study on five richer waveform families (baseline multitone, symmetric two-segment, echoed, complex-envelope, basis-expanded) to suppress residual-Z errors while targeting family-C (structured) and family-D (random) block-diagonal conditional rotations.

Results. Best result: baseline multitone on family-C, $N_{\text{active}} = 2$, $\chi T/2\pi = 3$, restricted process fidelity = 0.842, average gate fidelity = 0.873.

Critical weakness (confirmed by Study 4 rerun). The baseline multitone result on family-D ($N_{\text{active}} = 4$) achieves restricted process fidelity = 0.008 (essentially zero). Yet the single-ground-state probe fidelity is 0.574 and the spanning-quartet mean drops to 0.443. *This study's headline metric is the restricted process fidelity, which is completely misleading for the hard targets. The waveform is not implementing the intended operation.* Richer waveforms did not fix this; the study correctly notes that family-D targets are hard but does not escalate the failure to a primary conclusion.

3.3 Study 3: Ideal SQR Direct vs Echoed (`ideal_sqr_direct_vs_echoed_multitone`)

Goal. First study specifically targeting strict ideal x-axis SQR (all θ_n equal, all $\phi_n = 0$). Audit finding: earlier studies used arbitrary SU(2) block targets, not the strict x-axis target.

Results.

- Best direct: average gate fidelity 0.7245 on $N_{\text{active}} = 2$, $\chi T/2\pi = 5.0$ (chi_only model). Per-block fidelities are highly uneven: block 0 process fidelity 0.794, block 1 only 0.531. **Optimizer success flag: False.** The solution is at a poor local minimum.
- Echoed constructions: do not materially improve strict SQR fidelity. The mean strict process fidelity for echoed is lower than for direct on most cases.
- Study 4 multi-input validation for this study's best artifact gives joint process fidelity = 0.824, spanning-quartet mean = 0.875. The improvement over Study 2 is real but still far from 0.99.

Weakness. The convergence failure is acknowledged but not resolved. The study presents echoed results as comparable to direct without emphasizing that echoed constructions fail for strict SQR because the inserted X_π pulse is strongly manifold-dependent (see Study 8 diagnostics below).

3.4 Study 4: Multi-Input Validation (`native_multitone_sqr_fixed_multiinput_validation`)

Goal. Validate saved artifacts from Studies 1–3 using a three-tier probe ladder: single ground, selected pair (g, +x), and spanning quartet.

Results. This is the most critical audit study in the collection. Confirmed results from the rerun of `run_study.py` on 2026-03-28:

Source study	Construction	F_{joint}	Quartet mean	Quartet min	Assessment
Study 2	baseline multitone	0.0084	0.443	0.003	× fails
Study 5	direct multitone	0.601	0.734	0.346	∼ partial
Study 3	ideal SQR direct	0.824	0.875	0.720	∼ partial
Study 1	corrected unitary-opt	0.9986	1.000	1.000	✓ passes

The Study 2 case is especially striking: restricted process fidelity was reported as 0.008 (correctly low), but the single-ground-state fidelity was 0.574, creating the appearance of partial success that is not real. The unitary-optimized corrected-metric case (Study 1) is the only one that passes the multi-input validation to high fidelity.

3.5 Study 5: Arbitrary Fock-Conditional Rotations (`multitone_sqr_arbitrary_fock_condition`

Goal. Broader study with arbitrary SU(2) block targets (not just strict ideal SQR), covering $N_{\text{active}} \in \{2, 3, 4, 5\}$, durations $\chi T/2\pi \in \{1, 3, 5, 7\}$, and both chi-only and chi+χ' models. Single-pulse versus explicit echoed constructions.

Results. Mean process fidelity: 0.350 for single-pulse, 0.225 for echoed. Best single-pulse: family-A, $N_{\text{active}} = 2$, chi_only, $F_{\text{avg}} = 0.714$.

Weakness. The conclusion that “single pulse remains best overall” is correct in mean terms but undersells the depth of the failure. A mean of 0.35 represents mostly failed optimizations. The result that echoed is *worse* on average is important but not given a satisfying explanation; the study attributes it to waveform degrees of freedom without investigating whether the X_π quality is the primary cause.

3.6 Study 6: Strong Validation (`strong_validation_arbitrary_fock_conditional_rotations`)

Goal. Five waveform families (single-pulse Gaussian, independent-half echo, symmetry-constrained echo, segmented relaxed CPSQR, basis-expanded benchmark). 114 evaluations. Strict and relaxed objectives evaluated jointly.

Key results.

- Best strict case: single-pulse Gaussian, $F_{\text{strict}} = 0.9957$, $N_{\text{active}} = 2$, $\chi T/2\pi = 5.0$, chi_plus_chiprime, in-plane-axes target.
- Segmented relaxed CPSQR: achieves joint process fidelity ≈ 1.000 on hard targets ($N_{\text{active}} = 3$, random).
- Higher-expressivity basis-expanded benchmark: **zero improvement over baseline**.

Weakness. The zero gain from the basis-expanded family is suspicious and not investigated. It suggests either a broken optimizer path, an insufficient budget, or a genuine ansatz-vs-physics mismatch, none of which are resolved. The "best strict case" at 0.9957 is the headline result of this study, but it comes from the restricted view only and is not independently multi-input-validated within this study's own framework.

3.7 Study 7: Native/Rich Multitone Feasibility (`native_rich_multitone_sqr_cpsqr_feasibility`

Goal. Most comprehensive SQR study. Screens 10 waveform families in three stages: initial screen on chi+χ' model, comparison across both models and up to $N_{\text{active}} = 4$, and duration refinement sweep. Introduces the full joint process fidelity and the negative control recomputation.

Key findings (confirmed against saved data).

- **Best strict-SQR**: reduced_unitary_direct and native_direct_strict, $F_{\text{strict,joint}} = 0.9990$ on chi_only, $N_{\text{active}} = 2$, $\chi T/2\pi = 6.0$, smooth_x target. On $\chi T/2\pi = 5.0$: $F_{\text{strict,joint}} = 0.9988$. Both chi-only and chi+$\chi'$ models give the same best result.
- **Best CPSQR**: echoed_independent, $F_{\text{CPSQR,joint}} = 1.0000$ on chi_plus_chiprime, $N_{\text{active}} = 2$, $\chi T/2\pi = 5.0$, staggered_x target. Also achieves 1.0000 for chi_only and for $N_{\text{active}} = 3$.
- **Critical caveat**: On one case (echoed_asymmetric, chi_only, $N_{\text{active}} = 2$, staggered_x), the reduced quartet fidelity is 0.9999997 but the full joint process fidelity is only 0.8677. *The reduced-manifold view is not a safe proxy for joint fidelity.*
- **Performance scaling with N_{active}**: $N_{\text{active}} = 2$: best = 0.9990, mean = 0.721; $N_{\text{active}} = 3$: best = 0.9975, mean = 0.675; $N_{\text{active}} = 4$: best = 0.9948, mean = 0.646.
- Echoed families have *lower* mean strict fidelity than direct families for all N_{active}, confirming Study 5.
- The negative control recomputation showed that the old restricted metric (0.4949) underestimated performance under the patched cqed_sim (correct quartet = 0.875); metric inconsistency between studies is real.

Weakness. The study does not include a fresh GRAPE or unconstrained time-domain comparison on the hard cases, so it cannot distinguish ansatz-limited failure from physics-limited failure for the n_active ≥ 3 cases.

3.8 Study 8: Definitive Aggregation (`the_definitive_ideal_sqr_gate_study`)

Goal. Load and synthesize all four key source studies into a single machine-readable table. Add standalone diagnostics: X_π pulse characterization, spectral crowding analysis, parameter scaling, addressing window scaling fits.

Rerun (2026-03-28). The full rebuild completed in 36.98 s (wall clock). All four sanity checks pass:

- Baseline best direct: $F_{\text{strict}} = 0.6557$ (≈ 0.65 as expected)
- Native-rich strict > 0.99: value = 0.9990 ✓
- Native-rich CPSQR > 0.999: value = 1.0000 ✓

- Row counts: 64 (Study 3) + 270 (Study 5) + 348 (Study 7) + 240 (Study 2) = 922 total ✓

Weakness. The study is an aggregation pass only. No new optimizations were run. The "practical ranking" uses a decoherence-proxy score, not actual drift sweeps. The IMPROVEMENTS.md lists three P1 gaps: no full echoed-grid rerun with the compromise X_π pulse, no fresh drift analysis, and no GRAPE/unconstrained upper bound.

New diagnostics produced.

1. X_π **characterization**: The refocusing pulse (40 ns Gaussian, baseline parameters) degrades strongly with Fock level: level 0: $F = 0.9993$; level 1: $F = 0.9716$; level 2: $F = 0.8926$; level 3: $F = 0.7734$; level 4: $F = 0.6298$. A compromise pulse (shorter duration, adjusted sigma) improves the worst-manifold fidelity to 0.919 but does not certify echoed strict-SQR.

2. **Spectral crowding**: At $\chi T/2\pi = 3.0$, the tone bandwidth equals twice the manifold spacing (crowding ratio = 2.0) for all levels. This is a constant spectral bottleneck that does not worsen with manifold index.

3. **Parameter count**: Minimum free parameters for ideal SQR = $2N_{\text{active}}$. The current families use $6-18$ parameters for $N_{\text{active}} = 2$, giving overcomplete representations that the optimizer handles well.

3.9 Study 9: Holographic Cluster-State Synthesis (`cluster_state_holographic_unified`)

Goal. Implement two control problems using structured gate families $\mathrm{D}+\mathrm{R}+\mathrm{SQR}$ and $\mathrm{D}+\mathrm{R}+\mathrm{CPSQR}$: (a) the full joint logical transfer unitary $\mathrm{SWAP}\cdot\mathrm{CZ}\cdot(\mathrm{H}\otimes\mathrm{I})$ on $\{|g,0\rangle, |g,1\rangle, |e,0\rangle, |e,1\rangle\}$; and (b) the restricted ground-sector transfer $U_{\text{joint}}(|g\rangle \otimes |\psi\rangle) \approx |g\rangle \otimes H_c|\psi\rangle$ on $\{|g,0\rangle, |g,1\rangle\}$.

Critical finding. The full-target winners discovered in objective (a) *completely fail* on objective (b): their ground-sector fidelities collapse to ≈ 0.50 with ancilla excitation near 1. This is because those solutions route amplitude through the excited ancilla sector. The two objectives are genuinely different control problems.

Key ground-sector results (from saved data, confirmed consistent):

Family	Blocks	Ordering	$F_{\text{gnd},12}$	$\text{Ancilla}_{\text{worst}}$	$\text{Leakage}_{\text{worst}}$
D+R+SQR	2	SRD	0.9876	8.9×10^{-4}	2.97×10^{-2}
D+R+SQR	3	RDS	0.9997	4.7×10^{-4}	5.97×10^{-4}
D+R+SQR	4	RDS	0.9965	3.75×10^{-3}	6.96×10^{-3}
D+R+SQR	5	RDS	0.9999	4.1×10^{-5}	1.77×10^{-4}
D+R+CPSQR	1	CPRD	0.8615	≈ 0	2.59×10^{-1}
D+R+CPSQR	2	CPRD	0.9887	1.6×10^{-9}	3.05×10^{-2}
D+R+CPSQR	3	DRCP	0.9979	1.6×10^{-3}	4.20×10^{-3}
D+R+CPSQR	4	RDCP	0.9999	9.0×10^{-5}	1.76×10^{-4}
D+R+CPSQR	5	DRCP	0.9995	8.99×10^{-4}	1.12×10^{-3}

Both families first cross the 0.99 threshold at 3 blocks. The 2-block frontier (SQR: 0.988, CPSQR: 0.989) is just below threshold.

The GRAPE benchmark at $N_{\text{cav}} = 12$ for the *full target* reaches replay fidelity 0.976 at 800 ns. No like-for-like GRAPE baseline was run for the restricted ground-sector objective.

Weaknesses.

- The ground-sector follow-up used a warm-start from the best full-target solutions, not a fresh unrestricted search. A dedicated low-depth search could change the minimum-depth conclusion.
- No open-system validation for any of the ground-sector winners.
- The GRAPE frontier is a full-target benchmark only; comparing structured-family ground-sector fidelity to full-target GRAPE fidelity is not a like-for-like comparison.
- The non-monotonic depth dependence (4-block SQR $= 0.997 < $ 3-block SQR $= 0.9997$) reflects optimization variance, not a fundamental non-monotonicity.

4 Unified Methodology

4.1 Notation and Target Conventions (Unified)

Throughout this review, all fidelity claims use the **full joint process fidelity** $F_{\text{strict,joint}}$ for strict SQR and $F_{\text{CPSQR,joint}}$ for relaxed CPSQR, defined as:

$$F_{\text{joint}} = \frac{1}{d^2} \left| \text{tr} \left(U_{\text{target}}^{\dagger} U_{\text{optimized}} \right) \right|^2 \tag{4}$$

where the trace is over all d joint qubit-cavity logical levels. Reduced metrics (qubit-only, per-manifold, single-state) are shown as supplementary information only.

The targeted SQR target is:

$$U_{\text{target,strict}} = \bigotimes_{n=0}^{N_{\text{active}}-1} R_x(\pi/2) \otimes \mathbf{1}_{\text{unaddressed}} \tag{5}$$

(uniform $\pi/2$ rotation on all addressed manifolds for "smooth_x" targets; staggered angles for "staggered_x").

4.2 Waveform Families Compared

The following families appear across the studies. This table unifies naming conventions:

Internal family name	Display name	Parameters ($N_{\text{active}} = 2$)
`gaussian_seed`	direct seed	12
`native_direct_strict`	native direct	12
`reduced_unitary_direct`	reduced unitary	12
`symmetric_two_segment`	two-segment direct	18
`complex_envelope`	rich envelope	15
`basis_expanded`	basis-expanded	15
`echoed_independent`	independent echo	24
`echoed_asymmetric`	asymmetric echo	24
`echoed_symmetric`	symmetric echo	12
`echoed_cpsqr`	CPSQR echo	24
`segmented_relaxed`	segmented CPSQR	variable

5 Rerun Results and Reproducibility

5.1 What Was Rerun

1. **Definitive aggregation study** (`run_full_study.py --full`): Completed in 36.98 s on 2026-03-28. All sanity checks pass. Results match saved data exactly. **Reproducible.** ✓
2. **Multi-input validation study** (`run_study.py`): Completed on 2026-03-28. Results match saved data exactly (see Table in Section 3.4). **Reproducible.** ✓
3. **Native-rich key metrics** (read from saved JSON, spot-checked via manual queries): Best strict = 0.9990, best CPSQR = 1.0000, performance scaling with N_{active} confirmed. **Consistent with claims.** ✓
4. **Cluster-state ground-sector summary** (read from saved JSON): Both families cross 0.99 at 3 blocks. Best absolute: CPSQR 4-block = 0.9999. **Consistent with claims.** ✓

5.2 What Was Not Rerun

- The full native-rich 3-stage optimization pipeline (`run_study.py` for `native_rich`): computationally expensive, saved results are internally consistent, not rerun.
- The cluster-state design-space search (8-worker multiprocessing run): saved results are self-consistent, not rerun.
- The GRAPE frontier extension: computationally expensive, saved results used.
- Any fresh unconstrained optimization or GRAPE run for strict SQR (identified as the critical missing experiment in all IMPROVEMENTS.md files).

5.3 Metric Inconsistency Across Studies

A notable finding is that the "restricted process fidelity" computed in Studies 2–5 is systematically more optimistic than the full joint process fidelity computed in Studies 7–8. For the negative-control case (chi+χ', $N_{\text{active}} = 3$, $\chi T/2\pi = 3.0$, direct multitone):

- Reported restricted process fidelity (Study 3): $F = 0.4949$
- Recomputed joint process fidelity (Study 7): $F = 0.8242$
- Recomputed spanning-quartet mean (Study 7): $F = 0.875$

The direction of the discrepancy is counterintuitive: the full-system metric is *higher* than the old restricted metric. This reflects a cqed_sim API patch that corrected a phase convention between the two eras. **No quantitative result from Studies 2–5 can be directly compared to Studies 7–8 without metric normalization.**

6 Key Quantitative Results

6.1 Best Strict Ideal-SQR Results

Family	Model	N_{active}	$\chi T/2\pi$	$F_{\text{strict,joint}}$	$F_{\text{CPSQR,joint}}$
reduced_unitary_direct	chi_only	2	6.0	**0.9990**	0.9997
native_direct_strict	chi_only	2	6.0	**0.9990**	0.9997
reduced_unitary_direct	chi+χ'	2	6.0	**0.9990**	0.9997
native_direct_strict	chi+χ'	2	6.0	**0.9990**	0.9997
gaussian_seed	chi+χ'	2	5.0	0.9985	0.9999
reduced_unitary_direct	chi_only	3	5.0	0.9975	0.9994
reduced_unitary_direct	chi_only	4	(long)	0.9948	—
echoed_independent	chi_only	2	3.0	0.8671	**1.0000**
echoed_independent	chi+χ'	2	5.0	0.8677	**1.0000**
echoed_independent	chi_only	3	5.0	0.6784	**1.0000**
echoed_asymmetric	chi_only	3	3.0	0.6769	**1.0000**

Main observation: For strict SQR, direct families (reduced_unitary_direct, native_direct_strict) dominate at $N_{\text{active}} = 2$–4. Echoed families dominate for CPSQR but are worse for strict SQR because the inserted X_π is manifold-dependent.

6.2 Performance Scaling with Number of Addressed Manifolds

From the all_results.json of Study 7 (432 rows, all families, all $\chi T/2\pi$):

N_{active}	Best strict (any fam, any $\chi T/2\pi$)	Mean strict	Best CPSQR
2	0.9990	0.721	1.000
3	0.9975	0.675	1.000
4	0.9948	0.646	1.000

The *best* strict fidelity remains above 0.99 for all tested N_{active}, but only at long durations ($\chi T/2\pi \geq 3$–5) and only for specific (smooth, staggered) target families. The *mean* strict fidelity across all cases is 0.65–0.72, indicating that the 0.99 regime is narrow and not the typical outcome.

6.3 Minimum Duration for High Fidelity (Direct Families)

For native_direct_strict, chi_only model, smooth_x target:

N_{active}	Threshold	Minimum $\chi T/2\pi/2\pi$ achieving threshold
2	0.99	3.0
2	0.995	3.0
2 (CPSQR)	0.99	2.0

At $|\chi|/2\pi = 2.84\,\text{MHz}$, $\chi T/2\pi = 3$ corresponds to a pulse duration of $T = 3/(2\pi \times 2.84 \times 10^6\,\text{Hz}) \approx 168\,\text{ns}$.

6.4 X_π Pulse Quality Across Manifolds

The 40 ns Gaussian X_π pulse used in echoed constructions has sharply degrading quality with Fock level:

Fock level n	Process fidelity (baseline)	Process fidelity (robust 20 ns)
0	0.9993	0.9996
1	0.9716	0.9808
2	0.8926	0.9325
3	0.7734	0.8788
4	0.6298	0.7875

This is the primary reason echoed constructions fail for strict SQR when $N_{\textbf{active}} \geq 2$. Even the compromise pulse only improves the worst-manifold X_π quality to ≈ 0.919. A full echoed strict-SQR success would require a manifold-specific X_π sequence, which is itself a selective rotation problem.

7 Synthesis: What the Program Tells Us

7.1 The Core Physics Picture

The dispersive Hamiltonian encodes a per-manifold shift $\delta_n = -\chi n - \frac{\chi'}{2} n(n-1)$ on the qubit transition frequency. An ideal strict SQR at duration T requires the qubit to undergo a uniform rotation $\theta_n = \theta_0$ for all addressed n while accumulating zero net conditional phase. The phase accumulation rate is $\dot{\phi}_n = \delta_n$, which grows linearly with n.

A multitone ansatz addresses this by placing one drive tone per addressed manifold. The minimum condition for zero conditional phase is that each tone produces the correct transverse rotation while producing zero longitudinal (Z) rotation on its own manifold and zero cross-talk on other manifolds. This is achievable in principle at long enough durations (when spectral crowding is not limiting), which explains the experimental observation that $F_{\text{strict}} > 0.99$ is achievable at $\chi T/2\pi \geq 3$ for $N_{\text{active}} = 2$.

For larger N_{active}, the control problem becomes harder because more manifolds must be simultaneously addressed with narrow tones, and the spectral crowding ratio (always ≈ 2.0 for uniform $|\chi|$ spacing) does not improve with N_{active}.

7.2 Why Echoed Constructions Fail for Strict SQR

The toggle-frame argument for echoed cancellation of conditional phase assumes that the inserted X_π pulse is a perfect bit-flip on every addressed manifold. Because the dispersive Hamiltonian is manifold-dependent, no single-frequency X_π pulse can be perfect on all manifolds simultaneously. The X_π pulse quality degrades from 0.999 at level 0 to 0.630 at level 4 (see above), which means the toggling-frame cancellation argument breaks down precisely where it is most needed (higher photon numbers).

This finding, first identified clearly in Study 8 and now supported by the standalone X_π characterization, explains why:

1. Echoed constructions have *lower mean strict fidelity* than direct constructions (Studies 5, 6, 7, confirmed).

2. Echoed constructions have *near-perfect CPSQR fidelity*: the CPSQR target explicitly allows the residual Z phases that the imperfect X_π fails to cancel.

3. There is an inherent tension: achieving strict SQR via echoing requires a manifold-selective X_π, which is itself a harder form of SQR.

7.3 Relation to the Cluster-State Problem

The cluster-state study is partially decoupled from the SQR program: it uses SQR and CPSQR as primitive building blocks but optimizes the full structured sequence rather than characterizing the primitives in isolation. The key result — that both families first cross 0.99 at 3 blocks for the restricted ground-sector task — is obtained by optimizing the full sequence, not by composing pre-optimized SQR blocks.

The poor quality of the full-target solutions for the ground-sector task (ancilla excitation ≈ 1) is a structural finding that holds regardless of how well the individual SQR primitives work. It shows that the two objectives (full transfer vs ground-sector transfer) are genuinely orthogonal.

8 Final Conclusions

8.1 Claims That Are Robustly Supported

1. **Strict ideal x-axis SQR at $N_{\mathbf{active}} = 2$ with $\chi T/2\pi \geq 3$:** Both the chi_only and chi+χ' dispersive models admit strict ideal SQR solutions above $F = 0.99$ using the reduced_unitary_direct and native_direct_strict waveform families. Best observed: $F_{\text{strict,joint}} = 0.9990$ at $\chi T/2\pi = 6.0$, $N_{\text{active}} = 2$. This result was confirmed from the saved data and is consistent with the definitive study rebuild.

2. **Strict SQR above 0.99 requires long enough pulses:** At $N_{\text{active}} = 2$, $F > 0.99$ requires $\chi T/2\pi \geq 3.0$. Shorter pulses fall below this threshold.

3. **Relaxed CPSQR achieves near-unit fidelity for all tested $N_{\mathbf{active}}$:** The echoed_independent family achieves $F_{\text{CPSQR,joint}} = 1.0000$ for $N_{\text{active}} = 2$–3 at $\chi T/2\pi \geq 3$, on staggered_x targets. This is confirmed by Study 7 and is consistent with all related studies.

4. **Echoed constructions fail for strict SQR due to X_π manifold dependence:** The X_π refocusing pulse fails increasingly at higher Fock levels. This is the primary reason echoed constructions have lower strict fidelity than direct constructions. This finding is supported by both the averaged statistics and the explicit X_π characterization.

5. **Multi-input validation is required to certify a gate:** Study 4 confirms that solutions passing single-state probes or restricted process fidelity can fail catastrophically on superposition probes. The reduced/restricted metric can be misleading in either direction (too optimistic or, with a convention patch, too pessimistic).

6. **Cluster state ground-sector task requires ≥ 3 blocks:** Both D+R+SQR and D+R+CPSQR families first exceed $F = 0.99$ at 3 blocks under the restricted ground-sector objective. Best absolute fidelity: D+R+CPSQR at 4 blocks, $F_{\text{gnd},12} = 0.9999$.

7. **Full-target structured solutions completely fail the ground-sector task:** Old full-target winners have ancilla excitation near 1 under the ground-sector objective. The two problems are genuinely different.

8.2 Claims That Are Uncertain

1. **Performance at $N_{\text{active}} \geq 5$:** No study tested $N_{\text{active}} = 5$ with the joint metric. Study 5 included $N_{\text{active}} = 5$ but used an unreliable restricted metric. Whether strict SQR above 0.99 is achievable at $N_{\text{active}} = 5$ is unknown.

2. **Whether strict SQR failures are ansatz-limited or physics-limited:** At $N_{\text{active}} = 3$–4, the best strict results are 0.9948–0.9975 and only at specific (long) durations. Without a GRAPE or unconstrained upper bound, it is unknown whether the current direct families are near the physical limit or whether a better ansatz (e.g., GRAPE-optimized or time-domain free) would achieve significantly higher fidelity.

3. **Robustness to parameter drift:** The best solutions are numerically clean but have not been tested against systematic drift in χ, ω_q, or drive amplitude. The IMPROVEMENTS.md files list drift analysis as a P1 gap for Studies 7 and 8.

4. **Hardware relevance of the cluster-state result:** The cluster-state ground-sector winners have excellent closed-system fidelity but have not been validated with open-system (realistic decoherence) models. The GRAPE frontier open-system fidelity at 800 ns is 0.919, substantially below the closed-system replay of 0.976.

5. **Minimum depth for the 2-block cluster-state frontier:** Both families are just below 0.99 at 2 blocks (SQR: 0.988, CPSQR: 0.989). Whether a fresh low-depth search could cross 0.99 at 2 blocks is unresolved.

8.3 Claims That Are Not Supported

1. **"Strict SQR is generally achievable across the tested grid."** The mean strict process fidelity across all tested cases is 0.65–0.72, with the 0.99+ regime confined to $N_{\text{active}} = 2$ at long durations. This is not a general success.

2. **"Richer waveform families improve strict SQR over the baseline."** The complex-envelope and basis-expanded families do not improve over the native direct family for strict SQR (Study 7 comparison stage). The basis-expanded family provided zero gain even in Study 6.

3. **"Echoed constructions achieve strict SQR."** Echoed families have *lower* mean strict fidelity than direct families for all tested conditions. While echoed constructions reliably achieve CPSQR, they do not achieve strict SQR.

4. **Any robustness claims from the "practical ranking"** in the definitive study. That ranking uses a decoherence-proxy score only and has not been validated against actual noise or drift.

5. **"The 2-block cluster-state boundary can be crossed with existing families."** Current data show 2-block performance below 0.99 for both families. The IMPROVEMENTS.md explicitly calls for a fresh low-depth search rather than concluding that 3 blocks is the fundamental minimum.

9 Recommended Next Steps

In priority order:

1. **[P1] Run a GRAPE/unconstrained upper bound for strict SQR at $N_{\mathbf{active}} = 3$–4, $\chi T/2\pi = 3$–5.** This is the most critical missing experiment. Until an unconstrained comparison is available, the question of whether current failures are ansatz-limited or physics-limited cannot be answered.

2. **[P1] Perform parameter-drift sensitivity analysis on the best direct strict-SQR solutions.** Specifically: sweep $\chi \pm 5\%$, $\omega_q \pm 1\,\text{MHz}$, drive amplitude $\pm 5\%$ around the best $N_{\text{active}} = 2$ and $N_{\text{active}} = 3$ solutions.

3. **[P2] Run the full echoed-grid rerun with the compromise X_π pulse.** The standalone X_π characterization (Study 8) identified a shorter robust pulse with better manifold uniformity. Running the echoed constructions with this pulse is a well-defined next experiment.

4. **[P2] Run a fresh 2-block low-depth cluster-state search** directly for the ground-sector objective (not warm-started from full-target solutions).

5. **[P2] Open-system validation for the 3-block and best-absolute cluster-state ground-sector winners.** The closed-system $F > 0.999$ does not guarantee good hardware performance.

6. **[P3] Extend the SQR study to $N_{\mathbf{tr}} = 3$ transmon levels.** All current studies use $N_{\text{tr}} = 2$. The second excited transmon level introduces additional dispersive shifts that could affect the multitone selectivity.

10 Reproducibility Summary

Study	Rerun status	Conclusion
corrected_sqr metric	Data read, Study 4 rerun confirms key case	✓Consistent
parameterized waveform	Study 4 rerun reveals failure at $N_{\text{active}} = 4$	✓Consistent
ideal_sqr direct vs echoed	Study 4 rerun confirms 0.824 joint fidelity	✓Consistent
native_multitone validation	Full rerun 2026-03-28, results match exactly	✓Reproducible
multitone arbitrary fock	Data consistent with definition	✓Consistent
strong validation	Data consistent; best strict 0.9957 confirmed	✓Consistent
native rich multitone	Data read, key metrics confirmed from JSON	✓Consistent
the definitive study	Full rebuild rerun 2026-03-28 (37 s), all checks pass	✓Reproducible
cluster state unified	Data read, ground-sector summary confirmed	✓Consistent

All nine studies are internally consistent. The two studies where full reruns were performed (`native_multitone_sqr_fixed_multiinput_validation` and `the_definitive_ideal_sqr_gate_study`) reproduced exactly. Python 3.12 with `cqed-sim 0.1.0` was used for all reruns.

Appendix: Study Provenance and Code Locations

Study	Main script	Key data output
1	scripts/run_study.py	data/study_summary.json
2	scripts/run_study.py	data/study_results.json
3	scripts/run_study.py	data/study_results.json
4	scripts/run_study.py	data/study_summary.json
5	scripts/run_study.py	data/study_results.json
6	scripts/run_study.py	data/study_results.json
7	scripts/run_study.py	data/all_results.json
8	scripts/run_full_study.py	data/new_results/study_summary.json
9	scripts/run_design_space_study.py	data/ground_sector_followup_summary.json

The cqed_sim package is installed editably at `C:\Users\jl82323\Box\...cQED_simulation`. The current version (0.1.0) introduced a phase-convention correction relative to the version used in Studies 2–5; this caused the metric discrepancy documented in Section 5.3. All results quoted in this report use the current patched package.